



Name: _____

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Learning Objectives

- Find eigenvalues by solving the characteristic equation $\det(A - \lambda I) = 0$
- Compute eigenvectors by solving $(A - \lambda I)v = 0$
- Recognize repeated and complex eigenvalues
- Connect eigenvalues to trace, determinant, and diagonalization

For eigenvalues: factor the characteristic polynomial completely. For eigenvectors: row reduce $(A - \lambda I)$ and express solutions in terms of free variables.

1. Find the eigenvalues of A by solving $\det(A - \lambda I) = 0$.

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Answer: _____

2. Find the eigenvectors corresponding to each eigenvalue of A .

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

Answer: _____

3. Find the eigenvalues of A . Are they real or complex?

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Answer: _____

4. Find the eigenvalues and eigenvectors of A .

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

Answer: _____



5. Find all eigenvalues of A (hint: A is upper triangular).

$$A = \begin{bmatrix} 2 & 5 & 3 \\ 0 & -1 & 4 \\ 0 & 0 & 3 \end{bmatrix}$$

Answer: _____

6. Given $\lambda = 1$ is an eigenvalue of A, find the corresponding eigenvector.

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$$

Answer: _____

7. Use trace and determinant to find eigenvalues of A without expanding fully.

$$A = \begin{bmatrix} 5 & 2 \\ 1 & 4 \end{bmatrix}$$

Answer: _____

8. Determine if A is diagonalizable. If yes, find matrices P and D such that $A = PDP^{-1}$.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

Answer: _____

9. Given A has eigenvalues $\lambda_1 = 2$, $\lambda_2 = -1$, find the eigenvalues of A^3 .

$$A\vec{v} = \lambda\vec{v} \Rightarrow A^3\vec{v} = \lambda^3\vec{v}$$

Answer: _____

10. Find the characteristic polynomial $p(\lambda) = \det(A - \lambda I)$ and the eigenvalues.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & 4 \end{bmatrix}$$

Answer: _____





Problems 3–4: complex and repeated eigenvalues — connect to geometry. Problem 7: shows the trace/det shortcut.
Problem 10: companion matrix characteristic polynomial (advanced).

Solutions

1. Find the eigenvalues of A by solving $\det(A - \lambda I) = 0$.

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

$$\rightarrow \det(A - \lambda I) = \det(\begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}) = (4-\lambda)(3-\lambda) - 2 = 0.$$

$$\rightarrow \lambda^2 - 7\lambda + 10 = 0 \rightarrow (\lambda - 5)(\lambda - 2) = 0.$$

$$\rightarrow \text{Eigenvalues: } \lambda = 5 \text{ and } \lambda = 2.$$

Answer: $\lambda_1 = 5, \lambda_2 = 2$

2. Find the eigenvectors corresponding to each eigenvalue of A .

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

$$\rightarrow \text{Eigenvalues (upper triangular): } \lambda = 3, \lambda = 2 \text{ (diagonal entries).}$$

$$\rightarrow \text{For } \lambda = 3: (A - 3I)v = 0 \rightarrow \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} v = 0 \rightarrow v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

$$\rightarrow \text{For } \lambda = 2: (A - 2I)v = 0 \rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} v = 0 \rightarrow v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Answer: $\lambda = 3: \vec{v} = (1, 0)^T, \lambda = 2: \vec{v} = (-1, 1)^T$

3. Find the eigenvalues of A . Are they real or complex?

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\rightarrow \det(A - \lambda I) = (0 - \lambda)(0 - \lambda) - (-1)(1) = \lambda^2 + 1 = 0.$$

$$\rightarrow \lambda^2 = -1 \rightarrow \lambda = \pm i.$$

$$\rightarrow \text{Complex eigenvalues indicate } A \text{ is a rotation matrix (90°).}$$

Answer: $\lambda = \pm i$ (complex, no real eigenvalues)

4. Find the eigenvalues and eigenvectors of A .

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

$$\rightarrow \det(A - 2I) = \det(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}) = 0. \lambda = 2 \text{ (multiplicity 2).}$$

$$\rightarrow (A - 2I)v = 0: \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} v = 0 \rightarrow v_2 = 0, v_1 \text{ free.}$$

$$\rightarrow \text{Only one independent eigenvector: } v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \text{ (defective matrix)}$$

Answer: $\lambda = 2$ (repeated), $\vec{v} = (1, 0)^T$



5. Find all eigenvalues of A (hint: A is upper triangular).

$$A = \begin{bmatrix} 2 & 5 & 3 \\ 0 & -1 & 4 \\ 0 & 0 & 3 \end{bmatrix}$$

→ For a triangular matrix, eigenvalues = diagonal entries.

→ $\lambda_1 = 2, \lambda_2 = -1, \lambda_3 = 3$.

Answer: $\lambda_1 = 2, \lambda_2 = -1, \lambda_3 = 3$

6. Given $\lambda = 1$ is an eigenvalue of A, find the corresponding eigenvector.

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$$

→ $(A - I)v = 0$: $A - I = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 1 & -1 & 0 \end{bmatrix}$.

→ Row reduce: $\begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

→ x_1 free. $x_1 = x_1$. x_2 free from row 3 (already zero).

→ Eigenvectors: $v = t[1, 1, 0]^T$.

Answer: $\vec{v} = t(1, 1, 0)^T$

7. Use trace and determinant to find eigenvalues of A without expanding fully.

$$A = \begin{bmatrix} 5 & 2 \\ 1 & 4 \end{bmatrix}$$

→ $\text{tr}(A) = \lambda_1 + \lambda_2 = 5 + 4 = 9$.

→ $\det(A) = \lambda_1 \lambda_2 = 20 - 2 = 18$.

→ Characteristic eq: $\lambda^2 - 9\lambda + 18 = 0 \rightarrow (\lambda - 6)(\lambda - 3) = 0$.

→ $\lambda_1 = 6, \lambda_2 = 3$.

Answer: $\lambda_1 = 6, \lambda_2 = 3$

8. Determine if A is diagonalizable. If yes, find matrices P and D such that $A = PDP^{-1}$.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

→ A is already diagonal with eigenvalues $\lambda_1 = 1, \lambda_2 = 2$.

→ Standard basis vectors are eigenvectors: $e_1 = [1, 0]^T, e_2 = [0, 1]^T$.

→ $P = I$ (identity), $D = A$. So $A = I \cdot A \cdot I^{-1} = A$. ✓

Answer: A is diagonal: $P = I, D = A$

9. Given A has eigenvalues $\lambda_1 = 2, \lambda_2 = -1$, find the eigenvalues of A^3 .

$$A\vec{v} = \lambda\vec{v} \Rightarrow A^3\vec{v} = \lambda^3\vec{v}$$

→ If $Av = \lambda v$, then $A^2v = \lambda^2v$ and $A^3v = \lambda^3v$.

→ Eigenvalues of A^3 are λ^3 for each eigenvalue λ of A.

→ $2^3 = 8, (-1)^3 = -1$.

Answer: $\lambda_1^3 = 8, \lambda_2^3 = -1$



10. Find the characteristic polynomial $p(\lambda) = \det(A - \lambda I)$ and the eigenvalues.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & 4 \end{bmatrix}$$

→ $\det(A - \lambda I)$: expand along first column.

→ This is a companion matrix for $p(\lambda) = \lambda^3 - 4\lambda^2 + 5\lambda - 2$.

→ Try $\lambda=1$: $1-4+5-2=0$. ✓ Factor: $(\lambda-1)(\lambda^2-3\lambda+2) = (\lambda-1)^2(\lambda-2)$.

→ Eigenvalues: $\lambda=1$ (multiplicity 2), $\lambda=2$.

Answer: $p(\lambda) = -\lambda^3 + 4\lambda^2 - 5\lambda + 2$

